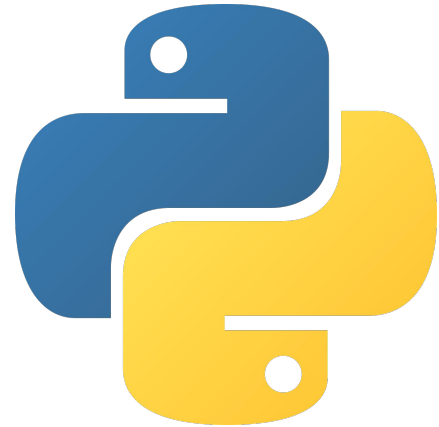




Python Week 4

Functions & Lists





Objective: Organizing your code and storing multiple pieces of data

What You'll Learn:

- What functions are
- How to create and call your own functions
- How to use lists to store multiple values



Do Now (Week 3 Review)

Write a program that:

1. Uses a for loop to count from 1 to 15.
2. If the number is divisible by 3, print "Fizz".
3. Otherwise, print the number.



```
for i in range(1, 16):  
    if i % 3 == 0:  
        print("Fizz")  
    else:  
        print(i)
```



I. Introduction to Functions



What is a function?

A function is a block of code that performs a specific task. Instead of writing the same code over and over, you can write it once and use it whenever you need it.



Why use functions?

- Reuse code without repeating yourself
- Keep your programs organized
- Make your code easier to read and understand



You've already used functions! These are built-in functions that Python provides. Today, you'll learn how to create your own!

```
print("Hello!")  
len("Python")  
input("What's your name? ")
```




True or False?

- A. A function lets you write code once and use it many times.
- B. You have to rewrite the same code every time you want to use it.
- C. `print()` is an example of a function in Python.



True or False?

- A. A function lets you write code once and use it many times.
a. Answer: True
- B. You have to rewrite the same code every time you want to use it.
a. Answer: False
- C. `print()` is an example of a function in Python.
a. Answer: True



II. How to Write Functions



Structure of a Function

```
def greet():  
    print("Hello!")
```

def => starts a function

greet => the name of the function

() => holds inputs (parameters)

print() => code that runs when the function is called



To use the function:

```
greet()
```

Calling a function => tells Python to run the code inside the function



Multiple Choice: What does the following code print out?

- A. 5 + 3
- B. 8
- C. 53
- D. Error

```
def add_numbers():  
    print(5 + 3)  
  
add_numbers()
```



Multiple Choice: What does the following code print out?

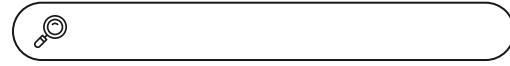
- A. 5 + 3
- B. 8**
- C. 53
- D. Error

```
def add_numbers():  
    print(5 + 3)  
  
add_numbers()
```



Try it: Write and call a function that prints out your name.

```
def greet():  
    print("Hello!")
```

```
def print_name():  
    print("Alex")  
  
print_name()
```



III. Lists



What is a list?

A list stores multiple values in one variable.

Why use lists?

- Store many pieces of data together
- Easily add, remove, or change items
- Use loops to go through every item



Instead of creating separate variables:

```
fruit1 = "Apple"  
fruit2 = "Banana"  
fruit3 = "Orange"
```

We can use a list:

```
fruits = ["Apple", "Banana", "Orange"]
```



List Structure

```
fruits = ["Apple", "Banana", "Orange"]
```

fruits => name of the list

[] => creates a list

"Apple", "Banana", "Orange" => items inside the list



Accessing Items

```
fruits = ["Apple", "Banana", "Orange"]  
  
print(fruits[0])
```

Output

```
Apple
```



Index => the position of an item in a list

Remember:

- Lists start counting at 0
- First item = index 0
- Second item = index 1
- Third item = index 2



Try it: Create a list of your favorite foods and print the list.

Challenge: Print only the first item in the list.



```
foods = ["Pizza", "Tacos", "Pasta"]  
  
print(foods)
```

```
print(foods[0])
```



Now it's your turn!

Write a Program that:

1. Creates a list of 3 favorite movies.
2. Uses a function to print each movie in the list.





Classwork Answer:

```
def print_movies(movie_list):  
    for movie in movie_list:  
        print(movie)  
  
movies = ["Cars", "Toy Story", "Finding Nemo"]  
  
print_movies(movies)
```



Thanks!

Have questions? Reach out:

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